9.1 Large-scale video classification with CNNs

- unlike images, which can be cropped and rescaled to a fixed size, videos vary widely in temporal extent and cannot be easily processed with a fixed-sized architecture
- thus, every video is treated as a bag of short, fixed-sized clips
- since each clip contains several contiguous frames in time, we can extend the connectivity of the network in the time dimension to learn spatio-temporal features
- there are multiple options for the precise details of the extended connectivity and we describe three broad connectivity pattern categories (Early Fusion, Late Fusion, and Slow Fusion) below
- afterwards, we describe a multiresolution architecture for addressing the computational efficiency

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- several approaches to fusing information across temporal domain are investigated (Figure 1):
  - the fusion can be done early in the network by modifying the first layer convolutional filters to extend in time, or
  - it can be done late by placing two separate single-frame networks some distance in time apart and fusing their outputs later in the processing
- we first describe a baseline single-frame CNN and then discuss its extensions in time according to different types of fusion

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- **Single Frame**: a Single Frame baseline architecture is used to understand the contribution of static appearance to the classification accuracy
- this network accepts inputs of size $170 \times 170 \times 3$ pixels
- using shorthand notation, the full architecture is $C(96, 11, 3)$-N-P-$C(256, 5, 1)$-N-P-$C(384, 3, 1)$-$C(384, 3, 1)$-$C(256, 3, 1)$-P-$FC(4096)$-$FC(4096)$, where $C(d, f, s)$ indicates a convolutional layer with d filters of spatial size $f \times f$, applied to the input with stride s, $FC(n)$ is a fully-connected layer with n nodes
- all pooling layers P pool spatially in non-overlapping 2×2 regions and all normalization layers N are defined as described in the AlexNet paper
- the final layer is connected to a softmax classifier with dense connections

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- **Early Fusion**: the Early Fusion extension combines information across an entire time window immediately on the pixel level
- this is implemented by modifying the filters on the first convolutional layer in the single-frame model by extending them to be of size $11 \times 11 \times 3 \times T$ pixels, where T is some temporal extent ($T = 10$ is used, or approximately a third of a second)
- the early and direct connectivity to pixel data allows the network to precisely detect local motion direction and speed

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- **Late Fusion**: the Late Fusion model places two separate single-frame networks (as described above), up to the last convolutional layer $C(256, 3, 1)$ with shared parameters a distance of 15 frames apart and then merges the two streams in the first fully-connected layer
- therefore, neither single-frame tower alone can detect any motion, but the first fully-connected layer can compute global motion characteristics by comparing outputs of both towers

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- **Slow Fusion**: the Slow Fusion model is a balanced mix between the two approaches that slowly fuses temporal information throughout the network, such that higher layers get access to progressively more global information in both spatial and temporal dimensions
- this is implemented by extending the connectivity of all convolutional layers in time and carrying out temporal convolutions in addition to spatial convolutions to compute activations
- in the used model, the first convolutional layer is extended to apply every filter of temporal extent $T = 4$ on an input clip of 10 frames through valid convolution with stride 2 and produces 4 responses in time
- the second and third layers above iterate this process with filters of temporal extent $T = 2$ and stride 2
- thus, the third convolutional layer has access to information across all 10 input frames

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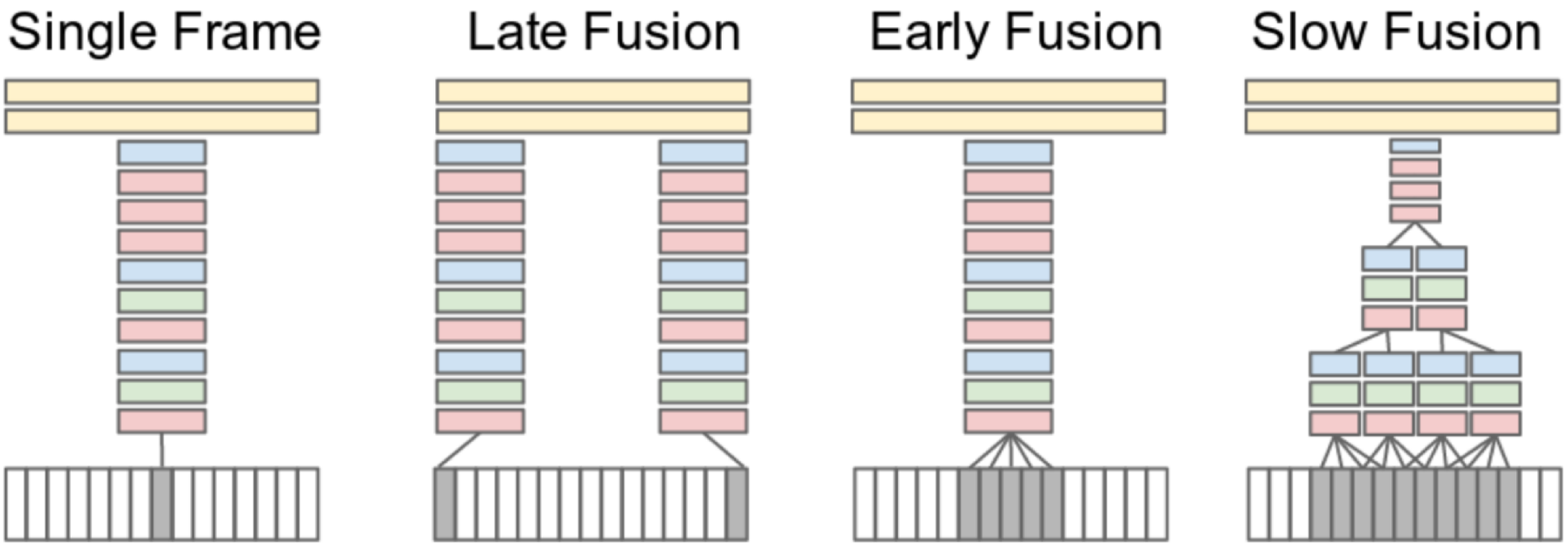


Figure 1: Explored approaches for fusing information over temporal dimension through the network. Red, green, and blue boxes indicate convolutional, normalization, and pooling layers, respectively. In the Slow Fusion model, the depicted columns share parameters.

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- **Multiresolution CNNs**: the runtime performance is a critical component to the ability to experiment with different architecture and hyperparameter settings
- this motivates approaches for speeding up the models, while still retaining their performance
- there are multiple fronts to these endeavors, including improvements in hardware, weight quantization schemes, better optimization algorithms, and initialization strategies, but we focus on changes in the architecture that enable faster running times without sacrificing performance

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- one approach to speeding up the networks is to reduce the number of layers and neurons in each layer, but this is found to consistently lower the performance
- instead of reducing the size of the network, further experiments were conducted on training with images of lower resolution
- however, while this improved the running time of the network, the high-frequency detail in the images proved critical to achieving good accuracy

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- **Fovea and context streams**: the proposed multiresolution architecture aims to strike a compromise by having two separate streams of processing over two spatial resolutions (Figure 2)
- a 178×178 frame video clip forms an input to the network
- the context stream receives the downsampled frames at half the original spatial resolution (89×89 pixels), while the fovea stream receives the center 89×89 region at the original resolution
- in this way, the total input dimensionality is halved
- notably, this design takes advantage of the camera bias present in many online videos, since the object of interest often occupies the center region

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- **Architecture changes**: both streams are processed by identical networks as the full-frame models, but starting at 89×89 clips of video
- since the input is only half the spatial size of the full-frame models, the last pooling layer is taken out to ensure that both streams still terminate in a layer of size $7 \times 7 \times 256$
- the activations from both streams are concatenated and fed into the first fully-connected layer with dense connections

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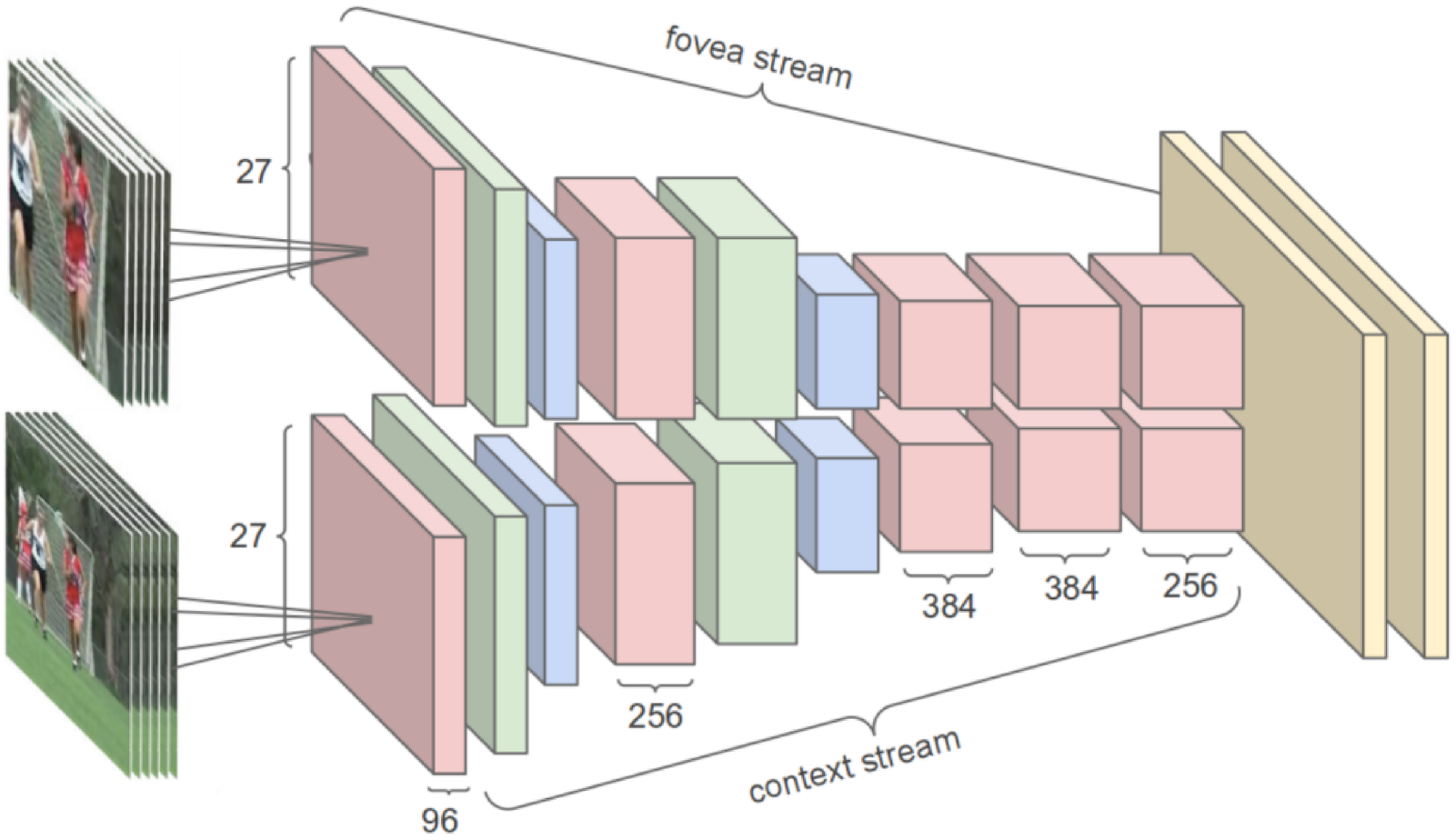


Figure 2: Multiresolution CNN architecture. Input frames are fed into two separate streams of processing: a *context stream* that models low-resolution image and a *fovea stream* that processes high-resolution center crop. Both streams consist of alternating convolution (red), normalization (green), and pooling (blue) layers. Both streams converge to two fully-connected layers (yellow).